



Alexander Dorsk

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GitHub: github.com/adorsk

Skills

- Programming Languages: Python, Javascript, Java, CoffeeScript, PHP, Ruby, Perl, C, C++, Go
- Libraries/Frameworks/Tools
 - Python: Flask, Django, SQLAlchemy, NumPy, Pandas
 - Javascript: React, D3, OpenLayers, jQuery, Webpack
 - DevOps: Docker, Puppet, Fabric, Vagrant
 - Databases: PostgreSQL, MariaDB/MySQL, SQLite, Redis, MongoDB
 - PHP: Drupal
 - Search/Analytics: Elasticsearch, BigQuery
- GIS: Mapserver, Mapnik, OpenLayers, GMT, GeoTools, GeoServer
- Cloud Services: Amazon services, Google Compute Engine
- Data Cleanup: OpenRefine

Experience

Software Engineer, MIT Digital Learning

March 2018 - May 2018

- Developed web application application for serving MIT lecture videos
- Frameworks: React, Django, Salt

Software Engineer, Aspuru-Guzik Lab, Harvard University

October 2016 - September 2017

- Designed and developed workflow orchestration framework.
- Trained researchers in the use of Jupyter notebooks and python.
- Helped researchers develop visualizations for computational chemistry data.
- Supervision of summer intern on database consolidation project.

Web Application Engineer, Spotify

October 2015 - September 2016

- Designed and developed web-based data curation tools.
- Developed ingest/cleanup pipelines and databases for music metadata and cultural data from web scrapers.
- Developed/deployed infrastructure via Docker, Puppet, Jenkins, GitHub Enterprise
- Supervision of summer interns on internal tool projects.

Instructor, Olin College of Engineering

September 2015 - December 2015

- Taught 1-credit undergraduate seminar on engineering & design in the context of musical instruments.
- Explored the history and philosophy of musical instrument design.
- Built instruments in hardware and software.

Software Consultant, Luminoso Technologies, Inc,
June 2015 - October 2015

- Developed javascript visualizations for semantic text analysis.
- Created React + D3 widget dashboard builder, using the Flux architecture.

Software Consultant, Addgene, Inc,
June 2014 - December 2014

- Migration of a legacy genetics database website to a DjangoCMS site.
- Coordination and training of 25+ employees in HTML and CMS use.
- Conversion of 500+ legacy HTML/Javascript pages to a new grid design.
- Refactoring of backend python code for improved performance and maintainability.

Founder/Engineer, Musikata
January 2014 - June 2015

- Development of games for online music education
- <https://github.com/musikata>

Software Consultant, Woods Hole Oceanographic Institute
February 2013 - June 2015

- Development of a Python toolkit for proteomics data analysis, for the [Saito Lab](#)
- <https://github.com/saitomics/metatryp>

GIS Consultant, Integrated Statistics/NOAA
January 2012 - February 2013

- Development of prototype web-based GIS visualization framework for analyzing fisheries model results.
- Standalone programs for running fisheries spatial planning models.
- <https://github.com/adorsk-noaa>

Data/Systems Librarian, MBLWHOI Library
January 2010 - January 2012

- Management of oceanographic data including research vessel data, deep submergence vehicle data and video, hydrographic data, and other data collections.
- Data rescue of historical oceanographic data and institutional archive

materials, including metadata reconstruction, data integration, digitization, and database creation.

- Metadata vocabulary and authority development and management.
- Development of tools and techniques for cleaning metadata including custom Drupal modules, Google Refine workflows, and custom scripting.
- Web design and systems administration including Drupal development, Chef infrastructure automation development, graphic design.
- Development of workflows for data access and publication
 - SCOR/IODE/MBLWHOI Workshop on Data Publication (<http://www.scor-int.org/Publications/wr230.pdf>)
 - Data Set to DOI using SWORD (<http://hdl.handle.net/1912/5080>)

Oceanographic Engineer, Woods Hole Oceanographic Institution

June 2007 - January 2010

- Resident technician on oceanographic research vessels for 30+ cruises, 500+ days at sea.
- Support of oceanographic research at sea including network administration, data management, maintenance/operation of satellite systems, mooring operations, and maintenance/operation of oceanographic equipment including CTD carousels, echosounders, ADCPs.
- Design and development of interactive shipboard data displays.
- R2R metadata project: design and development of EventLogger application.

Developer/Customer Support, Lux Scientiae

September 2004 - May 2007

- Software developer for email and web applications: Perl, web services, JavaScript.
- Customer support for a secure email and internet clients.

Computer Science Instructor, MIT MITES Summer Program

June 2004 - September 2004

- Teaching and tutoring computer science for high school students
- Residential counselor for 8-week residential program at MIT

Americorps Volunteer, City Year Boston

September 2002 - June 2003

- Created and taught social justice curriculum for middle school students.
- Database developer and administrator.
- Winner of 2003 Hale & Dorr Civic Innovation Award
- Press: '[Becoming Civic Superheroes](#)'

Research Activities

2015

- Saito, M. A., Dorsk, A., Post, A. F., McIlvin, M. R., Rappé, M. S., DiTullio, G. R. and Moran, D. M. (2015), Needles in the blue sea: Sub-species specificity in targeted protein biomarker analyses within the vast oceanic microbial metaproteome. *Proteomics*, 15: 3521–3531. doi:10.1002/pmic.201400630

2012

- Presentation: Devenish, Ann, Chandler, Cynthia L., Dorsk, Alexander, Raymond, Lisa, Work, Tobias, "Dataset to DOI using SWORD [poster]", Poster session, March 12, 2012, SPARC (The Scholarly Publishing and Academic Resources Coalition) Open Access Meeting, Kansas City, MO, March 12-13, 2012, <https://hdl.handle.net/1912/5080>

2010

- (Acknowledgements) The Final Case of the Decoding Delay Problem for Maximum Rate Complex Orthogonal Designs. Spence Adams, S; Karst, N.; Murugun, M. IEEE Transactions on Information Theory, V. 56, No. 1, Jan. 2010

2009

- Rolling Deck to Repository (R2R): A Research Cruise Prototype Event Logging System. Chandler, C.L.; Dorsk, A. AGU Fall 2009

2008

- Rolling Deck to Repository III: Shipboard Event Logging. Chandler, C.L.; Arko, R.; Miller, S.P.; Dorsk, A.; McCue, S.; Fisichella, D.C. AGU Fall 2008.

2006:

- Numerical Simulation of Sea Ice Dynamics with the Discrete Element Method. A. Dorsk, S. Ji. Clarkson Marine Science REU 2006. http://web2.clarkson.edu/projects/reushen/reu_china/Website/assets/participants_assets/Reports/Bing.pdf
- A. Dorsk. Transport and Structure of the NW Providence Channel. SEA Report C-204. <https://www.sea.edu/documents/cruisereports/C204cruisereport.pdf>

2005

- Coefficients of Friction of Non-newtonian fluids
 - Design of measuring apparatus
 - Data collection
 - B. S. Tilley, C. Laughlin, A. Dorsk, and Y. Zastavker, "On Puck Motion Down a Thin Film on an Inclined Plane", Society of Industrial and Applied Mathematics (SIAM), Boston, MA, July 2006

Education

Olin College of Engineering

BS, Engineering. 2003 - 2007

Concentration in Software Engineering

Four-year all-expenses scholarship.

Extracurriculars:

- Trombone, Olin Conductorless Orchestra
- Theatre: Franklin W. Olin Players.
- 'Engineering Discovery': Leader of an after-school program to introduce engineering to middle school students

Community

Strummerville Ukulele Club

Cambridge, Massachusetts. February 2015 - September 2017

- Mayor

Boston Music Technology Group

Boston, Massachusetts. June 2014 - June 2015

- Organizer

Other Skills & Interests

- Sailing: ARC 2017: Crossed Atlantic in Swan 47 sailboat, from Canaries to St. Lucia, 25 days.
- Music:
 - Instruments: Guitar, Ukulele ,Accordion, Piano, Trombone.
 - Composing Silent Film scores: [Zorro Score](#)
- Languages: Spanish, Brazilian Portuguese, Mandarin Chinese, French
- Photography: <http://www.flickr.com/photos/adorsk>
- Sports: Cycling, Swimming